

§ 195. On the class number in the real field contained in Ω_m .

Our general principles can also be applied in order to determine the class numbers in the divisors of the field Ω_m .

In particular, the formulae (6), (7), § 192 are applicable when we know the degrees of the prime ideals in such a divisor. The question concerning the ratio of the class numbers in the divisors of a field Ω_m to the class number in Ω_m itself is of great interest, and shall here be treated for the case in which the matter concerns the real field H_m contained in Ω_m , which we have already investigated in § 176 ff.¹ It has there resulted that q is, in the field H_m , the $\frac{1}{2}\mu$ -th power of a prime number of first degree, that the other prime numbers p decompose into e_1 prime factors of f_1 -th degree, so that

$$f_1 e_1 = \frac{1}{2}\mu, \quad (1)$$

and that two kinds of prime numbers p_1, p_2 are to be distinguished, so that for the prime numbers of the first kind $f_1 = \frac{1}{2}f$, $e_1 = e$, while for those of the second kind $f_1 = f$, $e_1 = \frac{1}{2}e$.

For the field H_m we accordingly obtain from § 192, (6), (7) the class number h_1 from the formula

$$g_1 h_1 = \lim_{s=1} (s-1) \Phi_1(s), \quad (2)$$

$$\Phi_1(s) = \frac{1}{1-q^{-s}} \prod_{p_1} \frac{1}{(1-p_1^{-\frac{1}{2}sf})^e} \prod_{p_2} \frac{1}{(1-p_2^{-sf})^{\frac{1}{2}e}}, \quad (3)$$

where the two product signs range over all prime numbers p_1 and p_2 . Using the symbols f_1, e_1 explained above, one may also write this as

$$\Phi_1(s) = \frac{1}{1-q^{-s}} \prod_p \frac{1}{(1-p^{-sf_1})^{e_1}}, \quad (4)$$

and extend the product sign over all prime numbers p .

The meaning of g_1 follows from § 190, (11).

We shall now single out from the group C of characters χ a divisor C_1 , which is to consist of all those characters χ_1 for which

$$\chi_1(-1) = +1. \quad (5)$$

It is first clear that these characters χ_1 form a group C_1 . This group is not identical with C , however, because there certainly is a character χ_0 for which $\chi_0(-1) = -1$. To obtain such a one, we need only put, in § 193, (6), a primitive μ -th root of unity for Θ , or put $\varepsilon = -1$ in § 193, (8). Then the totality of the characters $\chi_0 \chi_1$, which do not belong to C_1 , forms a coset C_2 , whose elements may be denoted by χ_2 . Every character χ is therefore contained either in C_1 or in C_2 ; for if χ does not occur in C_1 , then $\chi_0^{-1} \chi$ does occur in it, and consequently χ occurs in C_2 . Hence C_1 is a divisor of C of index 2.

The characters χ_1 may be formed in the following way.

1) If m is odd, then, as in § 193, we put

$$n \equiv c^\gamma \pmod{m},$$

and in

$$\chi_1(n) = \Theta^\gamma \quad (6)$$

¹Compare Kummer, "Bestimmung der Anzahl u.s.f.", Crelle's Journal, vol. 40 (1849).

let Θ run through the totality of the $\frac{1}{2}\mu$ -th roots of unity.

2) If m is even, then we put

$$n \equiv (-1)^{\alpha} 5^{\beta}$$

and obtain

$$\chi_1(n) = \Theta^{\beta}, \quad (7)$$

where Θ again runs through all the $\frac{1}{2}\mu$ -th roots of unity.

Both statements result easily from the fact that for $n = -1$ one has, in the first case, $\gamma = \frac{1}{2}\mu$, and in the second, $\alpha = 1, \beta = 0$, so that both times $\chi_1(-1) = 1$, and from the fact that each of the formulae (6), (7) represents exactly $\frac{1}{2}\mu$ different characters.

What is now essential is the following:

If p is any prime number different from q , then in the form $\chi_1(p)$, as χ_1 runs through the group C_1 , one obtains every f_1 -th root of unity, and every one equally often, namely e_1 times.

That all $\chi_1(p)$ are roots of unity of degree f_1 is clear, since always (compare § 177)

$$p^{f_1} \equiv \pm 1 \pmod{m},$$

and therefore

$$[\chi_1(p)]^{f_1} = \chi_1(\pm 1) = 1.$$

If however among the $\chi_1(p)$ a primitive f_1 -th root of unity occurs, then all other f_1 -th roots of unity also occur among them, and equally often, namely as often as the root of unity 1.

This follows immediately from the group nature of the χ_1 . Thus it remains only to show that among the $\chi_1(p)$ a primitive f_1 -th root of unity occurs.

This possibility follows very simply from the determination of χ_1 . First suppose that, in (6), with m odd, Θ is a primitive root of unity of degree $\frac{1}{2}\mu$, and that γ is the index of p . Then a power of Θ^{γ} , say $\Theta^{\gamma\lambda}$, is equal to 1 if and only if

$$2\gamma\lambda \equiv 0 \pmod{\mu}.$$

Now $\mu = ef$, and e is the greatest common divisor of γ and μ ; therefore this condition becomes $2\lambda \equiv 0 \pmod{f}$. But by § 177, 3, $f_1 = f$ or $f_1 = \frac{1}{2}f$ according as f is odd or even, and consequently one must have $\lambda \equiv 0 \pmod{f_1}$. Thus Θ^{γ} is a primitive f_1 -th root of unity.

If, however, m is even, we likewise take in (7) a primitive $\frac{1}{2}\mu$ -th root of unity for Θ , and put

$$p \equiv (-1)^{\alpha} 5^{\beta} \pmod{m},$$

so that f is the smallest positive number satisfying the congruences

$$\alpha f \equiv 0 \pmod{2}, \quad \beta f \equiv 0 \pmod{\frac{1}{2}\mu}.$$

Now $\Theta^{\beta\lambda}$ is equal to 1 only when $\beta\lambda \equiv 0 \pmod{\frac{1}{2}\mu}$. If $\lambda \neq 1$, so that $\beta \neq 0$, then the smallest value that λ can have is even, and hence this smallest value of λ is f .

But if $\lambda = 1$, then β must be 0, and two further cases are possible:

$$\begin{aligned} &\text{either } f = 1, \quad \alpha = 0, \quad p \equiv 1 \pmod{m}, \quad \lambda = f, \\ &\text{or } f = 2, \quad \alpha = 1, \quad p \equiv -1 \pmod{m}, \quad \lambda = \frac{1}{2}f. \end{aligned}$$

Only in this last case does p belong to the first kind, and it follows therefore that here also, in general, Θ^{β} is a primitive f_1 -th root of unity. The theorem is thereby proved.

This theorem permits us to apply formula § 193, (9), which here gives

$$(1 - p^{-sf_1})^{e_1} = \prod_{\chi_1} [1 - \chi_1(p)p^{-s}], \quad (8)$$

where the product is extended over all characters χ_1 of the group C_1 . Now the function $\Phi_1(s)$ may be developed further in the same way as $\Phi(s)$ in § 193. One obtains, on the same path as there, the formula (11),

$$\Phi_1(s) = \frac{1}{1 - q^{-s}} \prod_{\chi_1} \sum_{n=1}^{\infty} \frac{\chi_1(n)}{n^s}, \quad (9)$$

and when one passes to the limit $s = 1$,

$$g_1 h_1 = \prod_{\chi_1} \sum_{n=1}^{\infty} \frac{\chi_1(n)}{n}, \quad (10)$$

where now, however, the product is to be extended over all characters of the group C_1 with the exception of the principal character. This product is therefore a part of the product by which the number gh is expressed in § 193.

The numbers g, g_1 result from the general expression given in § 190, (11):

$$\frac{2^\nu \pi^{n-\nu} L}{w \sqrt{\pm \Delta}}. \quad (11)$$

Since, as was shown in § 178, the units in Ω_m and H_m , apart from the roots of unity, are the same, a fundamental system of units in H_m is at the same time a fundamental system in Ω_m .

Let therefore $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ denote a fundamental system of real units, and let $\varepsilon_{i,1}, \varepsilon_{i,2}, \dots, \varepsilon_{i,\nu}$ be the conjugate units to ε_i . Then the conjugate logarithms in the field H_m are

$$\log |\varepsilon_{i,1}|, \quad \log |\varepsilon_{i,2}|, \quad \dots, \quad \log |\varepsilon_{i,\nu}|,$$

whereas in the field Ω_m , in which only conjugate-imaginary pairs occur (§ 185), they are

$$2 \log |\varepsilon_{i,1}|, \quad 2 \log |\varepsilon_{i,2}|, \quad \dots, \quad 2 \log |\varepsilon_{i,\nu}|.$$

If therefore we denote by

$$E = \pm \begin{vmatrix} \log |\varepsilon_{1,1}| & \log |\varepsilon_{1,2}| & \cdots & \log |\varepsilon_{1,\nu-1}| \\ \log |\varepsilon_{2,1}| & \log |\varepsilon_{2,2}| & \cdots & \log |\varepsilon_{2,\nu-1}| \\ \cdots & \cdots & \cdots & \cdots \\ \log |\varepsilon_{\nu-1,1}| & \log |\varepsilon_{\nu-1,2}| & \cdots & \log |\varepsilon_{\nu-1,\nu-1}| \end{vmatrix} \quad (12)$$

the regulator of the field H_m (§ 187), then, in applying the expression (11), one must put in the field Ω_m

$$L = 2^{\nu-1} E,$$

and in the field H_m

$$L = E.$$

Further,

$$\begin{aligned} \text{in } \Omega_m : \quad n &= \mu, & \nu &= \frac{1}{2}\mu, \\ \text{in } H_m : \quad n &= \frac{1}{2}\mu, & \nu &= \frac{1}{2}\mu. \end{aligned}$$

In the field H_m only the two roots of unity ± 1 are contained. In Ω_m the positive and negative powers of r are contained, but no other roots of unity (§ 175), and in the enumeration it is

still to be noted that, when m is even, -1 is among the powers of r , while when m is odd it is not. Hence we have

$$\begin{aligned} \text{in } \Omega_m : \quad & w = 2m, \quad m \text{ odd}, \\ & w = m, \quad m \text{ even}, \\ \text{in } H_m : \quad & w = 2. \end{aligned}$$

Finally, if Δ, Δ_1 are the fundamental numbers of the fields Ω_m, H_m (§ 176), we still have

$$\pm\Delta = q\Delta_1^2 \quad (m \text{ odd}), \quad \pm\Delta = 4\Delta_1^2 \quad (m \text{ even}),$$

and it follows, if for simplicity we put ν for $\frac{1}{2}\mu$, that

$$\begin{aligned} g &= \frac{2^{2(\nu-1)}\pi^\nu E}{m\sqrt{q}\Delta_1}, & m \text{ odd}, \\ g &= \frac{2^{2(\nu-1)}\pi^\nu E}{m\Delta_1}, & m \text{ even}, \\ g_1 &= \frac{2^{\nu-1}E}{\sqrt{\Delta_1}}, \end{aligned} \tag{13}$$

whence

$$\begin{aligned} g &= \frac{2^{\nu-1}\pi^\nu g_1}{m\sqrt{q}\Delta_1}, & m \text{ odd}, \\ g &= \frac{2^{\nu-1}\pi^\nu g_1}{m\sqrt{\Delta_1}}, & m \text{ even}. \end{aligned} \tag{14}$$

If we combine formula (10) with formula § 193, (15), and use the notation introduced in § 194,

$$X_1 = \sum \frac{\chi_1(n)}{n}, \quad X_2 = \sum \frac{\chi_2(n)}{n},$$

then we obtain

$$gh = g_1 h_1 \prod X_2.$$

Replacing g by its value (14), we obtain for the class number

$$h = kh_1, \tag{15}$$

if

$$\begin{aligned} k &= \frac{m\sqrt{q}\Delta_1}{2^{\nu-1}\pi^\nu} \prod X_2, \quad m \text{ odd}, \\ &= \frac{m\sqrt{\Delta_1}}{2^{\nu-1}\pi^\nu} \prod X_2, \quad m \text{ even}, \end{aligned} \tag{16}$$

and where [by (10), (13)]

$$h_1 = \frac{\sqrt{\Delta_1}}{2^{\nu-1}E} \prod X_1 \tag{17}$$

is the class number of the field H_m . In the latter expression the product extends over all characters χ_1 of the group C_1 with the exception of the principal character.

That k is an integer, and therefore that h is a multiple of h_1 , will soon be shown. The numbers k and h_1 are called the first and second factor of the class number.

In the case in which

$$m = 2^\kappa$$

is a power of 2, we have by § 176, (7),

$$\Delta_1 = 2^{(\kappa-1)2^{\kappa-2}-1},$$

and if we insert this value, then in this case we may write the preceding formulae as

$$\begin{aligned} k &= \frac{\sqrt{2} 2^{(\kappa-3)2^{\kappa-3}} 2^{\kappa}}{\pi^{\nu}} \prod X_2, \\ h_1 &= \frac{\sqrt{2} 2^{(\kappa-3)2^{\kappa-3}}}{E} \prod X_1. \end{aligned} \tag{18}$$

§ 196. Class number in the field of the eighth roots of unity.

We shall first treat the case $m = 8$, $\mu = 4$, $\nu = 2$, partly to illustrate the preceding results and partly to gain a basis for the complete induction to be used later.

Here, apart from the principal character, we have only three characters, namely, if

$$n \equiv (-1)^{\alpha} 5^{\beta} \pmod{8},$$

then

$$\chi_1(n) = (-1)^{\beta}, \quad \chi_2(n) = (-1)^{\alpha}, \quad \chi_3(n) = (-1)^{\alpha+\beta}.$$

For $n = -1$ one has $\alpha = 1$, $\beta = 0$, and consequently χ_1 belongs to the first kind, χ_2, χ_3 to the second kind [§ 195, (5)]. For the four numbers $n = 1, 3, 5, 7$ the following values of these characters result:

	1	3	5	7
χ_1	+1	-1	-1	+1
χ_2	+1	-1	+1	-1
χ_3	+1	+1	-1	-1

From this one obtains the three functions f_1, f_2, f_3 [§ 194, (2)]:

$$\begin{aligned} f_1(x) &= x - x^3 - x^5 + x^7 = x(1 - x^2)(1 - x^4), \\ f_2(x) &= x - x^3 + x^5 - x^7 = x(1 - x^2)(1 + x^4), \\ f_3(x) &= x + x^3 - x^5 - x^7 = x(1 + x^2)(1 - x^4). \end{aligned}$$

For r we may put

$$r = \frac{1+i}{\sqrt{2}}, \quad r^2 = i, \quad r^3 = -\frac{1-i}{\sqrt{2}}, \quad r^4 = -1,$$

and from this it follows that

$$\begin{aligned} f_1(r) &= 2\sqrt{2}, & f_2(r) &= 0, & f_3(r) &= 2\sqrt{2}i, \\ f_1(r^2) &= 0, & f_2(r^2) &= 4i, & f_3(r^2) &= 0, \\ f_1(r^3) &= -2\sqrt{2}, & f_2(r^3) &= 0, & f_3(r^3) &= 2\sqrt{2}i. \end{aligned}$$

For $x = r^4$ all three functions $f(x)$ vanish, and for $x = r^5, r^6, r^7$ the function $f_1(x)$ receives the same values, while $f_2(x)$ and $f_3(x)$ receive the opposite values, as for $x = r^3, r^2, r$.

Consequently, from § 194, (12), with regard to the trigonometric formula

$$\sin \frac{3\pi}{8} = \cos \frac{\pi}{8},$$

one obtains

$$X_1 = -\frac{1}{\sqrt{2}} \log \tan \frac{\pi}{8}, \quad X_2 = -\frac{\pi}{4}, \quad X_3 = -\frac{\pi}{2\sqrt{2}}.$$

Now

$$\tau = \frac{r-1}{r^2(r+1)} = \tan \frac{\pi}{8} = \sqrt{2} - 1$$

is a unit of the field H_8 , which here is nothing other than the quadratic field arising from $\sqrt{2}$, and the units in it are therefore obtained from the solutions of Pell's equation

$$u^2 - 2v^2 = \pm 1$$

in the form $u + v\sqrt{2}$.

The smallest positive solution of this Pell equation is however $u = 1$, $v = 1$, and consequently, by Vol. I, § 128, all units in H_8 are found in the form $\pm(1 + \sqrt{2})^a$, where a is a positive or negative integral exponent. The unit τ results from this for $a = -1$, and hence all units are also contained in the form

$$\pm\tau^a.$$

Thus τ is a fundamental unit, and the determinant E occurring in the formulae of the preceding paragraph [§ 195, (12)] here becomes, since τ is a proper fraction,

$$E = -\log \tau.$$

From § 195, (15), (18), one therefore obtains

$$h_1 = 1, k = 1, \quad h = 1.$$

The field Ω_8 is therefore a one-class field, that is, one in which the decomposition of the integers into prime factors is possible according to the same laws as in the field of rational numbers.

At the same time it has also resulted that all units in the field H_8 are representable in the form $\pm\tau^a$. The conjugate values of the unit τ are however

$$\tau_1 = \tan \frac{\pi}{8}, \quad \tau_3 = -\tan \frac{3\pi}{8} = -\tau_1^{-1},$$

and therefore have different signs. We conclude from this for this case the theorem:

A unit in H_8 which, together with its conjugates, is of the same sign is, apart from the sign, the square of a unit.

§ 197. Recurrent calculation of the class number in the field Ω_m , when m is a power of 2.

For the general determination of the class number in the field Ω_m , under the assumption that m is any power of 2 and greater than 8, we apply a recurrent procedure. Thus let

$$m = 2^\kappa, \quad \mu = 2^{\kappa-1}, \quad \nu = 2^{\kappa-2}, \quad \nu' = 2^{\kappa-3}. \quad (1)$$

Beside the field Ω_m we consider the field Ω_μ , which is a divisor of Ω_m and which we shall here also denote by Ω'_m . Let h' be the class number in the field Ω'_m , that is, the number obtained from h when κ is replaced by $\kappa - 1$, and put

$$h = h'H. \quad (2)$$

If we presuppose h' as already known, then all that remains is the calculation of H , which, as will be shown, is an integer. Decomposing h and h' as in § 195, we put

$$h = kh_1, \quad h' = k'h'_1, \quad H = AB, \quad (3)$$

$$k = k'A, \quad h_1 = h'_1B, \quad (4)$$

where k' is obtained from k , and h'_1 from h_1 , by replacing $\varkappa$ by $\varkappa - 1$. In general, we shall now indicate by accents on letters that $\varkappa - 1$ is to be put in place of $\varkappa$.

By § 176, (7), one then has

$$\Delta_1 = 2^{(\varkappa-1)2^{\varkappa-2}-1}, \quad \Delta'_1 = 2^{(\varkappa-2)2^{\varkappa-3}-1},$$

and consequently

$$\Delta_1 = 2^{\varkappa 2^{\varkappa-3}} \Delta'_1. \quad (5)$$

If E has the meaning of § 195, (12), and E' is obtained from E when $\varkappa$ is replaced by $\varkappa - 1$, then we put further

$$E = E'D, \quad (6)$$

whereby D is defined as a non-zero positive number. As regards the sums

$$X = \sum \frac{\chi(n)}{n},$$

one must recall that $\chi(n)$ [by § 193, (8)] is defined by

$$n \equiv (-1)^{\alpha} 5^{\beta} \pmod{m}, \quad \chi(n) = (\pm 1)^{\alpha} \Theta^{\beta}, \quad (7)$$

where Θ denotes a ν -th root of unity.

Among the ν -th roots of unity the ν' -th roots of unity are also contained, and among the characters χ also the characters for the field Ω'_m . We distinguish, accordingly, primitive and imprimitive characters of the field Ω_m by designating as primitive the characters χ proper to the field Ω_m , that is, those in which Θ is a primitive ν -th root of unity.

Among the sums X all sums belonging to the field Ω'_m also occur:

$$X' = \sum \frac{\chi'(n)}{n}.$$

If now in (4) one inserts for k, k' and h_1, h'_1 the expressions formed in § 195, (16), (17), and then cancels the sums X' , one obtains

$$\begin{aligned} \pi^{\nu'} A &= 2 \cdot 2^{2^{\varkappa-4}(\varkappa-2)} \prod X_2, \\ DB &= 2^{2^{\varkappa-4}(\varkappa-2)} \prod X_1, \end{aligned} \quad (8)$$

where now the products $\prod$ extend only over the sums X_1, X_2 formed with the primitive characters χ_1, χ_2 .

In the expressions for the X given in § 194, (12),

$$\begin{aligned} X_1 &= -\frac{1}{2m} \sum_{s=1}^{m-1} f_1(r^s) \log \left(\sin \frac{s\pi}{m} \right)^2, \\ X_2 &= -\frac{i\pi}{m^2} \sum_{s=1}^{m-1} s f_2(r^s), \end{aligned} \quad (9)$$

considerable simplifications now enter under the assumption that m is a power of 2 and that χ_1, χ_2 are primitive characters.

The number $1 + \mu$ has the indices $\alpha = 0$, $\beta = \nu'$, and consequently by (7)

$$\chi(1 + \mu) = \Theta^{\nu'} = -1,$$

when χ is a primitive character. Further, for an odd n ,

$$n(1 + \mu) \equiv n + \mu \pmod{m},$$

and consequently

$$\chi(n + \mu) = -\chi(n).$$

If we now consider the function $f(r^s)$,

$$f(r^s) = \sum_n \chi(n) r^{sn}, \quad (10)$$

where n runs through the odd numbers of a complete residue system for the modulus m , then, since in (10) we may replace n under the summation sign by $n + \mu$, we obtain

$$f(r^s) = -r^{\mu s} \sum \chi(n) r^{sn},$$

and since $r^\mu = -1$,

$$f(r^s) = -(-1)^s f(r^s).$$

Thus

$$f(r^s) = 0, \quad \text{if } s \text{ is even,} \quad (11)$$

and hence in formulae (9) we may restrict the summation letter s to the odd numbers which are smaller than m .

But from (10) one has

$$f(r^s) = \chi(s)^{-1} \sum_n \chi(ns) r^{sn},$$

and, replacing sn by n , which is permitted for odd s ,

$$f(r^s) = \chi(s)^{-1} f(r), \quad \text{if } s \text{ is odd.} \quad (12)$$

The function $f(r)$, if α, β are the indices of n , has the expression

$$f(r) = \sum (\pm 1)^\alpha \Theta^\beta r^n, \quad (13)$$

and is therefore equivalent to the cyclotomic resolvent considered in § 17 of this volume,

$$(\pm 1, \Theta, r),$$

for which in § 17, (17), the relation was established:

$$(\pm 1, \Theta, r)(\pm 1, \Theta^{-1}, r) = \pm m. \quad (14)$$

Here throughout the upper signs apply to the sums X_1 , the lower to the sums X_2 [§ 195, (7)].

We now divide the series of numbers s occurring in (9) into four parts. Let t be a symbol which runs through the series of positive odd numbers from 1 to $\nu - 1$; then the odd s runs through the four number series

$$t, \quad t + \mu, \quad m - t, \quad m - t - \mu,$$

and one has

$$\begin{aligned}\chi(t + \mu) &= -\chi(t), \\ \chi_1(m - t) &= \chi_1(t), & \chi_1(m - \mu - t) &= -\chi_1(t), \\ \chi_2(m - t) &= -\chi_2(t), & \chi_2(m - \mu - t) &= \chi_2(t).\end{aligned}$$

From (12) it follows that

$$\begin{aligned}f(r^{t+\mu}) &= -f(r^t), \\ f_1(r^{m-t}) &= f_1(r^t), & f_1(r^{m-\mu-t}) &= -f_1(r^t), \\ f_2(r^{m-t}) &= -f_2(r^t), & f_2(r^{m-\mu-t}) &= f_2(r^t).\end{aligned}$$

Dividing the sums (9) accordingly into four parts each and also using the trigonometric equation

$$\sin \frac{(t + \mu)\pi}{m} = \cos \frac{t\pi}{m},$$

one obtains

$$X_1 = -\frac{1}{m} \sum_t f_1(r^t) \log \left(\tan \frac{t\pi}{m} \right)^2, \quad X_2 = \frac{i\pi}{m} \sum_t f_2(r^t).$$

In this sum one has

$$\frac{t\pi}{m} < \frac{\pi}{4}, \quad \tan \frac{t\pi}{m} > 0,$$

and consequently, with regard to (12), (13), if we put Θ^{-1} for Θ , whereby $\chi(t)^{-1}$ passes into $\chi(t)$, we obtain

$$X_1 = -\frac{2}{m} (+1, \Theta^{-1}, r) \sum_{1 \leq t \leq \nu-1} \chi_1(t) \log \tan \frac{t\pi}{m}, \quad (15)$$

$$X_2 = \frac{i\pi}{m} (-1, \Theta^{-1}, r) \sum_{1 \leq t \leq \nu-1} \chi_2(t). \quad (16)$$

§ 198. The class-number factor A .

To calculate first A , we have to insert the expression (16) for X_2 into the first formula (8) of the preceding paragraph. This formula can also be represented as

$$\pi^{\nu'} A = 2m^{1/4} 2^{-\nu'} \prod X_2. \quad (1)$$

If we denote by α, β the indices of t , then $\chi_2(t) = (-1)^\alpha \Theta^\beta$;

$$\sum_t \chi_2(t) = \sum (-1)^\alpha \Theta^\beta = \varphi(\Theta) \quad (2)$$

is an integer of the field Ω_ν , and it becomes

$$X_2 = \frac{i\pi}{m} (-1, \Theta^{-1}, r) \varphi(\Theta).$$

Now for Θ all roots of the equation

$$\Theta^{\nu'} + 1 = 0 \quad (3)$$

are to be put, and for A one obtains, with regard to § 197, (14),

$$A = 2^{1-\nu'} \prod_{\Theta} \varphi(\Theta). \quad (4)$$

The sum by which the number $\varphi(\Theta)$ is defined in (2) contains ν' terms of the form $(-1)^\alpha \Theta^\beta$, where α, β are to be determined so that

$$(-1)^\alpha 5^\beta \equiv t \pmod{m} \quad (5)$$

and t lies between 0 and ν . First it is clear that among the exponents β no two can occur which are congruent modulo ν' . For if $\beta \equiv \beta' \pmod{\nu'}$, then $5^\beta \equiv 5^{\beta'} \pmod{\mu}$, and from

$$(-1)^\alpha 5^\beta \equiv t, \quad (-1)^{\alpha'} 5^{\beta'} \equiv t' \pmod{m}$$

it follows that

$$(-1)^\alpha t - (-1)^{\alpha'} t' \equiv 0 \pmod{\mu}.$$

Since however t and t' are absolutely smaller than $\nu = \frac{1}{2}\mu$, this is possible only when $t = t'$, $\alpha = \alpha'$. Thus β in (2) runs through a complete residue system modulo ν' .

But since β is to be taken modulo ν , the β determined by (5) will come out partly smaller and partly greater than ν' . If, then, we let β_1 run through the series of numbers $0, 1, \dots, \nu' - 1$, then we obtain in (2) two kinds of terms:

$$\begin{aligned} 1. \quad & (-1)^\alpha \Theta^\beta = (-1)^\alpha \Theta^{\beta_1}, \quad \beta = \beta_1 < \nu', \\ 2. \quad & (-1)^\alpha \Theta^\beta = -(-1)^\alpha \Theta^{\beta_1}, \quad \beta = \beta_1 + \nu' \geq \nu'. \end{aligned} \quad (6)$$

Now by (5) the absolutely smallest residue of $(-1)^\alpha 5^\beta$ modulo m is positive; therefore if we determine a number $c_\beta = \pm 1$ so that the absolutely smallest residue of $c_\beta 5^\beta$ is positive, then in case (6), 1,

$$c_{\beta_1} = (-1)^\alpha.$$

Further, $5^{\nu'} \equiv 1 \pmod{\mu}$, and since $5^{\nu'}$ is not also congruent to 1 modulo m , it follows that

$$5^{\nu'} \equiv 1 + \mu \pmod{m}. \quad (7)$$

Consequently, in case (6), 2,

$$(-1)^\alpha 5^\beta = (-1)^\alpha 5^{\beta_1} 5^{\nu'} \equiv (-1)^\alpha 5^{\beta_1} + \mu \pmod{m},$$

and hence here the absolutely smallest residue of $(-1)^\alpha 5^{\beta_1}$ modulo m is negative. Thus in case (6), 2,

$$c_{\beta_1} = -(-1)^\alpha$$

is to be put.

From this, by (2), it follows, if we again write β for β_1 ,

$$\varphi(\Theta) = \sum_{\beta=0}^{\nu'-1} c_\beta \Theta^\beta = c_0 + c_1 \Theta + c_2 \Theta^2 + \dots + c_{\nu'-1} \Theta^{\nu'-1}, \quad (8)$$

and here

$$c_\beta = +1 \quad \text{or} \quad = -1, \quad (9)$$

according as the absolutely smallest residue of 5^β is positive or negative. Thus $\varphi(\Theta)$ is an integer of the field Ω_ν .

It remains to determine the behaviour of this number toward the number 2, or rather toward the prime factor $(1 - \Theta)$ of 2 contained in Ω_ν . For this purpose we form

$$(1 - \Theta)\varphi(\Theta) = 2\psi(\Theta), \quad (10)$$

and this gives

$$\psi(\Theta) = \frac{c_0 + c_{\nu'-1}}{2} + \frac{c_1 - c_0}{2}\Theta + \frac{c_2 - c_1}{2}\Theta^2 + \dots + \frac{c_{\nu'-1} - c_{\nu'-2}}{2}\Theta^{\nu'-1}. \quad (11)$$

The coefficients in this expression are all 0 or 1, and hence $\psi(\Theta)$ also is an integer of the field Ω_ν . But for this number one obtains, putting $\Theta \equiv 1$,

$$\psi(\Theta) \equiv c_{\nu'-1} \equiv \pm 1 \pmod{(1 - \Theta)}, \quad (12)$$

and consequently $\psi(\Theta)$ is relatively prime to 2.

The product occurring in (4) is nothing other than the norm, taken with respect to the field Ω_ν , of the number $\varphi(\Theta)$; and if we denote this norm by N , then by § 169

$$N(1 - \Theta) = 2.$$

Thus by (10)

$$N\varphi(\Theta) = 2^{\nu'-1}N\psi(\Theta).$$

It follows that

$$A = N\psi(\Theta), \quad (13)$$

and hence, with regard to (12), that A is an odd integer.

If we denote by $A_\varkappa$ the number A belonging to $m = 2^\varkappa$, then from § 197, (4), by complete induction, the expression for the first factor of the class number is

$$k = A_4 A_5 \cdots A_\varkappa, \quad (14)$$

and from this the theorem follows:

1. The first factor of the class number is an odd integer.

The number $A_\varkappa$ is not difficult to calculate from formula (10) for the first values of $\varkappa$.

For $m = 16$ one finds $\nu' = 2$, $\beta = 0, 1$, $c_0 = c_1 = 1$, consequently

$$\psi(\Theta) = 1, \quad A_4 = 1.$$

For $m = 32$, $\nu' = 4$, $\beta = 0, 1, 2, 3$,

$$c_0 = 1, \quad c_1 = 1, \quad c_2 = -1, \quad c_3 = -1,$$

$$\psi(\Theta) = -\Theta^2,$$

so that also $A_5 = 1$.

For $m = 64$, $\nu' = 8$, $\beta = 0, 1, 2, 3, 4, 5, 6, 7$, the absolutely smallest residues of 5^β are

$$1, 5, 25, -3, -15, -11, 9, -19;$$

thus the c_β are

$$+1, +1, +1, -1, -1, -1, +1, -1,$$

and

$$\psi(\Theta) = -\Theta^3 + \Theta^6 - \Theta^7, \quad \psi(\Theta)\psi(-\Theta) = \Theta^6(\Theta^6 - 2i),$$

from which it follows easily that

$$A_6 = 17.$$

A somewhat longer calculation gives still

$$A_7 = 21121.$$

§ 199. The class-number factor B .

The class-number factor B must be calculated in an entirely different way. Here, if Θ is again a primitive root of the equation § 198, (3), namely $\Theta^{\nu'} + 1 = 0$, and

$$t \equiv (-1)^{\alpha} 5^{\beta} \pmod{m}, \quad (1)$$

then one is to put

$$\chi_1(t) = \Theta^{\beta}, \quad (2)$$

and formula § 197, (15) gives

$$X_1 = -\frac{2}{m}(1, \Theta^{-1}, r) \sum_t \Theta^{\beta} \log \tan \frac{t\pi}{m}. \quad (3)$$

Now for every odd t , if [with regard to (1)]

$$r = e^{2\pi i/m}, \quad r^{\nu} = i, \quad r^{t\nu} = (-1)^{\alpha} i,$$

one has

$$\tan \frac{t\pi}{m} = (-1)^{\alpha} r^t \frac{1 - r^t}{1 + r^t}, \quad (4)$$

and this, as we already saw in § 178, is a unit of the real field H_m . The conjugate values of this unit with respect to this field are obtained by putting $t \equiv 5^{\beta} \pmod{m}$ and by letting β run through a complete residue system modulo μ . We put accordingly

$$\tau_{\beta} = \tan \frac{5^{\beta}\pi}{m}. \quad (5)$$

By § 198, (7), $5^{\beta+\nu'} \equiv 5^{\beta} + \mu \pmod{m}$, and from this, by a known trigonometric formula, there follows

$$\tau_{\beta+\nu'} = -\frac{1}{\tau_{\beta}}. \quad (6)$$

If we therefore put

$$\lambda_{\beta} = \frac{1}{2} \log \tau_{\beta}^2, \quad (7)$$

then by (6)

$$\lambda_{\beta+\nu'} = -\lambda_{\beta}, \quad (8)$$

and therefore

$$\Theta^{\beta+\nu'} \lambda_{\beta+\nu'} = \Theta^{\beta} \lambda_{\beta}. \quad (9)$$

Among the values of β occurring in the sum (3), as we already saw in the preceding paragraph, no two are congruent modulo ν' , and if therefore by means of formula (9) we reduce all values of β to the interval from 0 to $\nu' - 1$, then we obtain all numbers of this interval.

Accordingly formula (3) may be represented as

$$X_1 = -\frac{2}{m}(1, \Theta^{-1}, r) \sum_{\beta=0}^{\nu'-1} \Theta^{\beta} \lambda_{\beta}. \quad (10)$$

This must be inserted in the expression given in § 197, (8),

$$DB = 2^{2\kappa-4(\kappa-2)} \prod X_1 = m^{\nu'} 2^{-\nu'} \prod X_1,$$

and then, if we put

$$U_{\Theta} = \sum_{\beta=0}^{\nu'-1} \Theta^{\beta} \lambda_{\beta}, \quad (11)$$

we obtain, with regard to § 197, (14),

$$DB = \prod_{\Theta} U_{\Theta}. \quad (12)$$

This product may now be represented in the following way by a determinant which contains only the conjugate logarithms of the units τ .

If we omit the index Θ in U , we may set up the system of equations

$$\begin{aligned} U &= \lambda_0 + \lambda_1 \Theta + \lambda_2 \Theta^2 + \cdots + \lambda_{\nu'-1} \Theta^{\nu'-1}, \\ \Theta U &= -\lambda_{\nu'-1} + \lambda_0 \Theta + \lambda_1 \Theta^2 + \cdots + \lambda_{\nu'-2} \Theta^{\nu'-1}, \\ &\dots \\ \Theta^{\nu'-1} U &= -\lambda_1 - \lambda_2 \Theta - \lambda_3 \Theta^2 + \cdots + \lambda_0 \Theta^{\nu'-1}, \end{aligned}$$

from which, by elimination of the powers of Θ , it follows that

$$\begin{vmatrix} \lambda_0 - U & \lambda_1 & \lambda_2 & \cdots & \lambda_{\nu'-1} \\ -\lambda_{\nu'-1} & \lambda_0 - U & \lambda_1 & \cdots & \lambda_{\nu'-2} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ -\lambda_1 & -\lambda_2 & -\lambda_3 & \cdots & \lambda_0 - U \end{vmatrix} = 0. \quad (13)$$

This is an equation of degree ν' for U , whose roots are the quantities U_{Θ} . If one orders it according to powers of U , then, since ν' is even, the highest power $U^{\nu'}$ of U receives the coefficient 1, and one therefore obtains the product of the roots occurring in (12), when $U = 0$ is put, from the determinant (13).

We shall arrange the determinant so obtained somewhat differently, leaving the first row in its place and writing the remaining rows in the reverse order and with changed signs. In doing this $\frac{1}{2}\nu'$ changes of sign are necessary, and therefore for the product (12) we obtain the determinant defined by the following equation, whose absolute value we denote by T :

$$(-1)^{\frac{1}{2}\nu'} \begin{vmatrix} \lambda_0 & \lambda_1 & \cdots & \lambda_{\nu'-2} & \lambda_{\nu'-1} \\ \lambda_1 & \lambda_2 & \cdots & \lambda_{\nu'-1} & -\lambda_0 \\ \lambda_2 & \lambda_3 & \cdots & -\lambda_0 & -\lambda_1 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \lambda_{\nu'-1} & -\lambda_0 & \cdots & -\lambda_{\nu'-3} & -\lambda_{\nu'-2} \end{vmatrix} = \pm T, \quad (14)$$

where to the right of the diagonal row running from left below to right above the negative signs stand.

The determinant T is here ordered in such a way that each row arises from the preceding one by the substitution $(r, r^{5^{\beta}})$. Each column therefore contains a system of conjugate logarithms.

By (12) one now has

$$B = \frac{T}{D}. \quad (15)$$

To obtain clarity concerning the nature of this result, it is necessary to consider somewhat more closely the number D which was defined by § 195, (12), § 197, (6); and this requires entering into the theory of the units of the field H_m .